

Zero-shot Detection of AI-Generated Image via RAW–RGB Alignment

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Abstract

Advances in generative AI (GenAI) have increasingly complicated the identification of synthetic images, prompting the proposal of numerous zero-/few-shot detection methods to counter unknown GenAI better. However, we observe that existing detectors often misclassify synthetic images with physical transformations (e.g., print+scan) as real. The essence of this observation lies in: should images remapped from the physical world to digital space still be categorized as “Synthetic”? Furthermore, the definition of what constitutes real and synthetic images urgently needs to be clarified. We first boldly propose that the authenticity of an image depends on whether it originates from the physical world, i.e., it is necessary to verify the original correlation between the digital image and the physical world. To this end, we first analyze the physical-to-digital mapping process: illumination signals are captured by camera sensors as RAW data, which is then converted into RGB data via camera internal parameters. This process embodies unique physical cues inherent to real scenes. Based on this, we propose a novel forensic feature termed alignment trace, which is constructed by modeling a shared RAW–RGB feature space. This trace captures the inherent parameter correlations of real images in the physical-to-digital conversion process, thereby indirectly verifying the physical origin of the image. Experiments demonstrate that our method achieves state-of-the-art zero-shot detection using only real RAW–RGB data pairs. When additional prior knowledge is provided, the method can be easily fine-tuned to achieve better cross-domain detection performance. We hope this work provides a new baseline for zero-shot synthetic detection and, more significantly, inspires the forensics community to explore the essential distinctions between real and synthetic images.

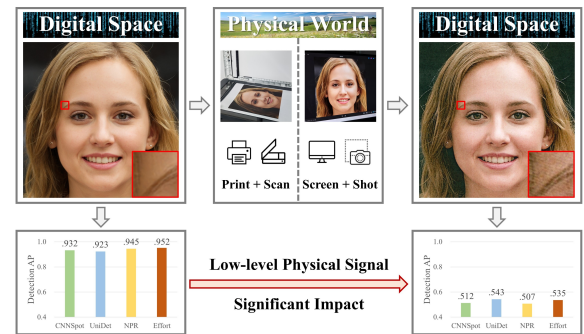


Figure 1. **Observation I:** Existing detectors [41, 52, 55, 68] often misclassify physical remapped synthetic images as real ones.

1. Introduction

GenAI, represented by GANs [19] and diffusion models [22], has advanced rapidly. While boosting innovation in entertainment, malicious AI-generated content (AIGC) can trigger cognitive mislead and trust crisis [33, 35]. To address this problem, numerous forensics [17, 27, 28, 38, 55, 60, 61, 63, 64], especially those based on zero-/few-shot paradigms, have been proposed in recent years to better defend against the emerging new GenAI.

Despite the certain successes made by these existing works, they face challenges in identifying synthetic images that have gone through physical remapping, e.g., “print+scan” or “screen+shot” (see Fig. 1). The core issue behind this limitation is the lack of a rigorous **definition for “Synthetic”**: should images that undergo physical remapping still qualify as “Synthetic”? To resolve this ambiguity, this work puts forward a clear definition: a synthetic image is exclusively an image that has no prior existence in the physical world and is created directly within the digital space. Per this definition, a synthetic image transitions into a real one after physical remapping, which explains why it can reasonably evade detection.

Under the proposed definition, the criterion for detecting real/synthetic images lies in whether they have/lack a prior association with the physical world. For real images,

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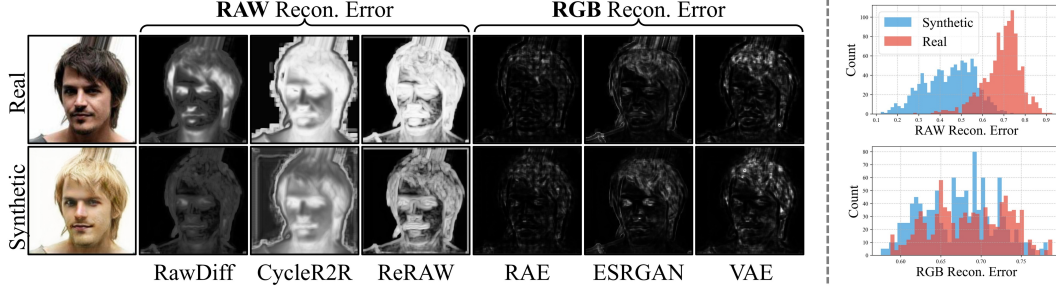


Figure 2. **Observation II:** Real images exhibit larger reconstruction errors than synthetic ones, especially with RAW-based reconstruction methods. Left: Qualitative results of three RAW methods (RawDiff [45], CycleR2R [36], ReRAW [8]) and three RGB methods (RAE [71], ESRGAN [56], VAE [26]). Right: Statistical quantitative results (comparing ReRAW and VAE) from 2000 real and synthetic images.

their generation process involves light intensity signals being captured as RAW signals by camera sensors, followed by conversion to RGB signals via an in-camera Image Signal Processor (ISP) [67]. In contrast, synthetic images are RGB signals generated directly in the digital space, with no underlying light or RAW signal sources. This naturally leads us to propose a detection method by checking if a test image has a RAW signal source that complies with physical laws. To validate this idea, we introduce RGB-to-RAW pipelines [8, 36, 45] to reconstruct the test RGB into RAW, and visualize the reconstruction errors in Fig. 2. For comparison, we additionally include the results of standard reconstruction, *i.e.*, RGB-only autoencoders [26, 56, 71]. Clearly, RAW-based reconstruction errors enable easy distinction between real and synthetic images, verifying that RAW signals serve as a strong clue to an image’s physical-world origin.

After preliminary verification, this work proposes a novel forensic feature termed **alignment trace**, designed to extract reliable clues for the existence of RAW signals embedded within RGB signals. The key attributes of this trace are three-fold: (1) **GenAI prior independence**: massive RAW-**RGB** pairs are automatically generated from only RAW data to enable a self-supervised training paradigm; (2) **RAW and RGB representation**: the utilization of topological graphs and the design of Stratified Entropy Attention (SEA) achieve more precise representation of RAW and RGB signals; (3) **RAW-**RGB** alignment**: Jointly optimized via RGB pixels, RAW topologies, and RAW pixels, enabling it to well characterize the properties of the RAW-to-**RGB** pipeline;

The three key attributes endow the proposed alignment trace with strong zero-shot discriminability. For instance, in the unsupervised detection of MidJourney synthesis, we achieve a clustering metric of 0.990 in Normalized Mutual Information (NMI), which is far higher than the 0.730 NMI of unsupervised competitor MIB [9]. Furthermore, this trace can be efficiently fine-tuned based on GenAI prior data to obtain enhanced few-shot discriminability. Specifically, for 10 types of cross-domain GenAI in the ForenSynths [55] dataset, we achieve an average Average Precision (AP) of

0.987, which outperforms the 0.953 AP of supervised state-of-the-art competitor Effort [68].

In summary, our main contributions are:

- Proposing a new definition of synthetic images bounded by the physical world.
- Revealing that RAW signals, as an effective bridge between the physical and digital spaces, can be used to distinguish real and synthetic images.
- Developing the alignment trace, which is based on an optimized joint space supervised by RGB pixels, RAW topologies, and RAW pixels.
- Demonstrating the effectiveness of the proposed trace through comprehensive zero-/few-shot evaluations.

2. Related Works

To address the evolving GenAI, numerous works [9, 11, 16, 20, 39, 41, 49, 52, 55, 59, 64, 68] have attempted to explore more universal forensic features from the perspectives of few-/zero-shot learning. From the few-shot perspective, Wang *et al.* [55] designed data augmentation methods and found that the traces of specific GANs can be effectively transferred to detect other GANs. Sha *et al.* [49], on the other hand, revealed that when specific diffusion models reconstruct images, there exist significant reconstruction errors between real and synthetic images. From the zero-shot perspective, many works aim to build implicit forensic models for real images without relying on GenAI priors, leveraging aspects such as statistical microstructures [39], color distribution [11], and lossless encoding [16]. Despite the considerable progress achieved by these works, most of them fail to provide a clear definition of what constitutes a synthetic image. Consequently, their evaluation criteria are limited to traces in the digital space, with little focus on the physical world.

3. Methodology

Fig. 3 presents the proposed method for modeling a feature space that aligns RAW-**RGB** signals. It first outlines data

generation and trace definition in Sec. 3.1, followed by an explanation of how such traces are jointly optimized from representations of RGB pixels in Sec. 3.2, RAW topologies in Sec. 3.3, and RAW pixels in Sec. 3.4. For the usage of zero-/few-shot detection, please refer to Appx. B.

3.1. Preparation

Data Generation. Let $\mathcal{R} = \{\mathbf{R}_i\}_{i=1}^I$ be a dataset comprising I raw image samples, where $\mathbf{R}_i$ is the i -th Raw instance. We define a set $\mathcal{P} = \{P_j\}_{j=1}^J$ consisting of J distinct RAW-to-RGB image processing pipelines, where each $P_j : \text{RAW} \rightarrow \text{RGB}$ represents the j -th entity. Consequently, by applying all pipelines of $\mathcal{P}$ to any member of $\mathcal{R}$, we can construct a comprehensive dataset $\mathcal{I} = \{\mathbf{I}_{i,j} | \mathbf{R}_i \in \mathcal{R}, P_j \in \mathcal{P}\}$, where $\mathbf{I}_{i,j} = P_j(\mathbf{R}_i)$. It is noticed that any P_j incorporates core operations from an ISP, such as demosaicing, white balance, color correction, and tone mapping, alongside conventional post-processing operations including compression, scaling, blurring, and noise addition. These operations can be implemented using different algorithms (e.g., AHD, DCB, DHT for demosaicing [2, 69]) or parameterizations (e.g., quality factor ranging from 50 to 99 for compression). Crucially, each P must include demosaicing and at most one instance each of white balance and tone mapping, with post-processing operations defined to have a maximum of 3 steps. This results in over 1000 distinct ISP combinations and $\sum_{i=1}^3 192^i$ post-processing combinations, yielding a total of $J = 1000 \times \sum_{i=1}^3 192^i \approx 10^9$ unique pipelines. Detailed specifications are provided in Appx. A.

Trace Definition. With the Data Generation, we define a *trainable latent feature extractor* $f : \mathcal{I} \rightarrow \mathbb{R}^d$, where d is the dimension of the feature space. For any $\mathbf{I}_{i,j} \in \mathcal{I}$, the extracted latent feature is $\mathbf{F}_j = f(\mathbf{I}_{i,j}) \in \mathbb{R}^d$. Notably, the subscript j in $\mathbf{F}_j$ emphasizes that this latent feature is expected to be solely dependent on the P_j rather than the $\mathbf{R}_i$. Ideally, with $\forall \mathbf{R}_i, \mathbf{R}_a \in \mathcal{R}, i \neq b$, and $\forall P_j, P_k \in \mathcal{P}, j \neq k$, we expect that $\mathbf{F}_j$ possesses following properties:

1. **Content Invariance:** $f(P_j(\mathbf{R}_i)) = f(P_j(\mathbf{R}_b))$;
2. **Discriminability:** $f(P_j(\mathbf{R}_i)) \neq f(P_k(\mathbf{R}_i))$.

According to these properties, for a query RGB image $\mathbf{I}_q \notin \mathcal{I}$, meaning there is no pair $(\mathbf{R}_i, P_j)$ from $\mathcal{R} \times \mathcal{P}$ such that $\mathbf{I}_q = P_j(\mathbf{R}_i)$, we can obtain the def. 3.1.

Definition 3.1 (Feature Deviation) For a predefined threshold τ , KL Divergence $\mathcal{D}_{KL}$ between $f(\mathbf{I}_q)$ and $f(\mathbf{I})$, $\forall \mathbf{I} \in \mathcal{I}$ satisfies:

$$\min_{\mathbf{I} \in \mathcal{I}} \mathcal{D}_{KL}(f(\mathbf{I}_q) || f(\mathbf{I})) \geq \tau. \quad (1)$$

This deviation indicates characteristics of $\mathbf{I}_q$ are generated by an unknown or non-standard RAW-to-RGB process, which fall outside the learned manifold or cluster boundaries

established by the diverse, yet finite, processing pipelines in $\mathcal{P}$ applied to the known RAW dataset $\mathcal{R}$.

3.2. RGB-Vision Branch

We now describe the extraction of $\mathbf{F}_j$ from $\mathbf{I}_{i,j}$. Instead of using the extractor f trained from scratch, to avoid overfitting and inherits rich visual priors efficiently, we propose to distill features from the pre-trained feature space of a frozen large vision model (LVM) $f_{\text{LVM}} : \mathcal{I} \rightarrow \mathbb{R}^{K \times d}$, where K is number of spatial tokens, and extract the intermediate feature $\mathbf{H} = (\mathbf{h}_1, \mathbf{h}_2, \dots, \mathbf{h}_K)$ as:

$$\mathbf{H} = f_{\text{LVM}}(\mathbf{I}_{i,j}). \quad (2)$$

Then, to achieve the effective feature distillation, we adapt a multi-level entropy-aware method, called Stratified Entropy Attention (SEA) $f_{\text{SEA}} : \mathbf{H} \rightarrow \mathbb{R}^d$, to extract $\mathbf{F}_j$ following:

$$\mathbf{F}_j = f_{\text{SEA}}(\mathbf{F}_j^0, \mathbf{H}), \quad (3)$$

where $\mathbf{F}_j^0$ is the latent feature randomly initialized. The purpose of f_{SEA} is to filter out information semantically related to content from $\mathbf{H}$ and instead focus more on information related to physics, specifically that associated with the physical-to-digital conversion. To this end, we introduce characteristics related to the source of $\mathbf{H}$ (i.e., pixels), such as regional smoothness and brightness. Given that entropy quantifies the uncertainty or disorder in pixel distribution, which is inherently linked to the physical state of the original scene, we calculate the aforementioned characteristics based on the concept of entropy. Furthermore, we employ stratified sampling to mitigate the impacts of extreme pixel value skews and uneven intensity distributions [8].

Specifically, f_{SEA} consists of entropy binning, stratified sampling, and attention fusion as follows.

Entropy Binning. Let $\pi_k \in \mathbb{R}^{s \times s \times 3}$ denote the pixel region (i.e., an $s \times s$ patch within $\mathbf{I}_{i,j}$) from which the high-level feature $\mathbf{h}_k$ is primarily extracted, where $k \in [K]$. We define the physical entropy $E_k \in [0, 1]$ of $\mathbf{h}_k$ using gradient entropy and luminance-chrominance (LC) entropy derived from π_k as:

$$E_k = \frac{1}{2} (\text{Grad}(\pi_k) + \text{LC}(\pi_k)). \quad (4)$$

Here, the $\text{Grad}(\cdot)$ measures local smoothness deviations:

$$\text{Grad}(\pi_k) = \frac{\text{Std}(\nabla \pi_k)}{\max_{\nabla m} \text{Std}(\nabla \pi_m)}, \quad (5)$$

where $\nabla \pi_k = \left\{ \sqrt{G_x(p, q)^2 + G_y(p, q)^2} \right\}$ represents the set of gradient magnitudes at each pixel $(p, q) \in [s] \times [s]$, with $G_x(p, q)$ and $G_y(p, q)$ denoting x and y directional gradients computed via Sobel operators [24]. The $\text{LC}(\cdot)$ captures

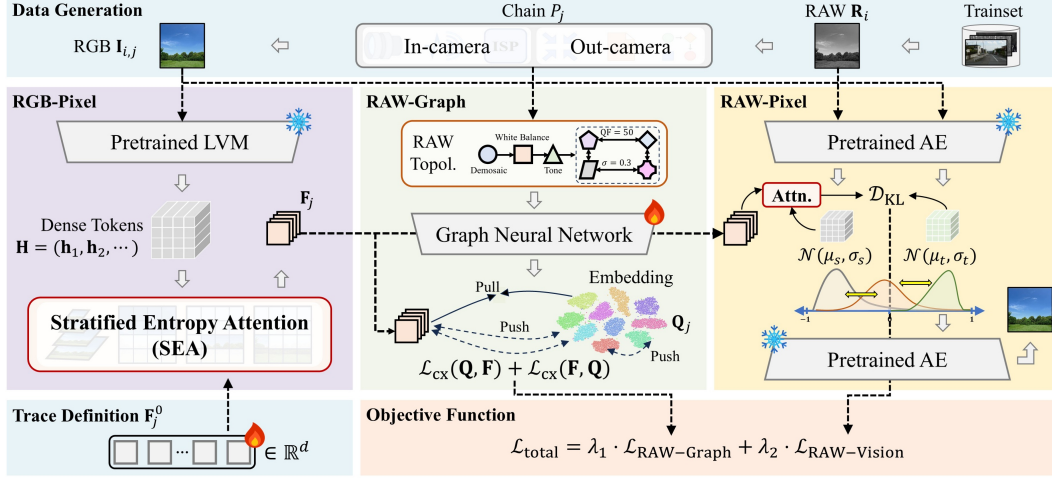


Figure 3. The overview of our proposed training scheme. After data generation and trace definition (Sec. 3.1), joint feature space alignment is performed through three branches: RGB pixels (Sec. 3.2), RAW topological graphs (Sec. 3.3), and RAW pixels (Sec. 3.4).

disrupted luminance-hue correlations:

$$LC(\pi_k) = \frac{\mathcal{J}(L_k, C_k) / \sqrt{\mathcal{H}(L_k) \cdot \mathcal{H}(C_k)}}{\max_{\forall m} \left(\mathcal{J}(L_m, C_m) / \sqrt{\mathcal{H}(L_m) \cdot \mathcal{H}(C_m)} \right)}, \quad (6)$$

where L_k and C_k are the luminance and hue sets from the YCrCb and HSV conversion of each pixel in π_k , with $\mathcal{H}(\cdot)$ denoting Shannon entropy and $\mathcal{J}(\cdot, \cdot)$ joint entropy [50, 62].

With these entropy values computed, we split all K tokens into B equidistant entropy bins $\mathcal{B}^b = [(b-1)/B, b/B)$ based on E_k , forming token sets $\mathcal{T}^b = \{\mathbf{h}_k \mid E_k \in \mathcal{B}^b\}$.

Stratified Sampling. To ensure balanced representation across all entropy levels, we uniformly select M tokens from every token set $\mathcal{T}^b$. This results in B sub-subsets $\hat{\mathcal{T}}^b = \{\hat{\mathbf{h}}_{b,m}\}_{m=1}^M$ (one per bin), which are concatenated to form a comprehensive representative subset of $\mathbf{H}$: $\hat{\mathcal{T}} = \bigcup_{b=1}^B \hat{\mathcal{T}}^b$. This subset $\hat{\mathcal{T}}$ retains proportional coverage of all entropy levels, ensuring no single range of physical consistency dominates the subsequent processing.

Attention Fusion. The aggregated subset $\hat{\mathcal{T}}$ from stratified sampling is used to refine the initial anchor feature $\mathbf{F}_j^0$ through attention mechanisms that prioritize tokens with stronger alignment to physical conversion laws. Specifically, $\mathbf{F}_j^0$ serves as the query to compute normalized attention weights over the tokens in $\hat{\mathcal{T}}$, with these weights explicitly amplifying contributions from tokens with varying entropy:

$$w_{b,m} = \frac{\exp\left(\frac{(1-E_{\hat{\mathbf{h}}_{b,m}}) \cdot \mathbf{F}_j^0 \cdot \hat{\mathbf{h}}_{b,m}^\top}{\sqrt{d}}\right)}{\sum_{b'=1}^B \sum_{m'=1}^M \exp\left(\frac{(1-E_{\hat{\mathbf{h}}_{b',m'}}) \cdot \mathbf{F}_j^0 \cdot \hat{\mathbf{h}}_{b',m'}^\top}{\sqrt{d}}\right)}, \quad (7)$$

where $E_{\hat{\mathbf{h}}_{b,m}}$ denotes the physical entropy of the token $\hat{\mathbf{h}}_{b,m} \in \hat{\mathcal{T}}^b$. The refined anchor feature is then generated by

fusing all tokens in $\hat{\mathcal{T}}$ using these weighted attention scores:

$$\mathbf{F}_j = \sum_{b=1}^B \sum_{m=1}^M w_{b,m} \cdot \text{Attn}(\mathbf{F}_j^0, \hat{\mathbf{h}}_{b,m}, \hat{\mathbf{h}}_{b,m}), \quad (8)$$

where

$$\text{Attn}(\mathbf{F}_j^0, \hat{\mathbf{h}}_{b,m}, \hat{\mathbf{h}}_{b,m}) = \text{SoftMax}(\mathbf{F}_j^0 \hat{\mathbf{h}}_{b,m}^\top / \sqrt{d}) \hat{\mathbf{h}}_{b,m} \quad (9)$$

represents scaled dot-product attention [54].

At this point, the refined $\mathbf{F}_j$ has incorporated full-spectrum signals from the RGB perspective, awaiting subsequent joint alignment from the RAW perspective.

3.3. RAW-Graph Branch

We now describe how to represent the RAW signal for alignment with the extracted feature $\mathbf{F}_j$ from the RGB signal. Unlike directly using RAW pixels as alignment constraints, we first propose incorporating parametric correlations involved in the RAW-to-RGB process as supplementary conditions, *i.e.*, reusing the operation chain $\mathcal{P}_j$ from the data generation stage. To this end, we encode the $\mathcal{P}_j$ as a topological graph and map it to the feature space using a Graph Neural Network (GNN) [65]. Our graph-based mapping differs from text-based work [72] primarily because text struggles to capture hierarchical dependencies and dynamic interaction logic between parameters, such as the sequential dependency between white balance and tone mapping in ISP processing. In contrast, graph structures can encode parameter features via node attributes, and represent logic and correlation strength through edge weights and directions, better aligning with the physical process of RAW-RGB conversion.

We formally define mapping from operation chain $\mathcal{P}$ to directed graph space as $\mathcal{P} \rightarrow \mathcal{Q}$, where $\mathcal{Q} = \{\mathcal{Q}_j\}_{j=1}^J$.

Specifically, for $Q \in \mathcal{Q}$, $Q = \{\mathcal{V}, \mathcal{E}, \omega\}$, where $\mathcal{V}$, $\mathcal{E}$, and ω respectively represent the vertex set, edge set, and edge weight function derived from the corresponding $P \in \mathcal{P}$. In details, $\mathcal{V} = \mathcal{V}_{\text{DM}} \cup \mathcal{V}_{\text{WB}} \cup \mathcal{V}_{\text{TM}} \cup \mathcal{V}_{\text{Post}}$ encompasses both ISP core operations (demaicing $\mathcal{V}_{\text{DM}}$, white balance $\mathcal{V}_{\text{WB}}$, and tone mapping $\mathcal{V}_{\text{TM}}$) and post-processing operations $\mathcal{V}_{\text{Post}}$. The size of each set corresponds to the number of distinct implementation algorithms for that operation type, *e.g.*, $|\mathcal{V}_{\text{DM}}| = 4$ indicates that demaicing is realized through 4 different algorithms. The directed edge set $\mathcal{E}$ enforces sequential constraints: all edges originate from $\mathcal{V}_{\text{DM}}$ as mandatory initial nodes, followed by optional edges to $\mathcal{V}_{\text{WB}}$ or $\mathcal{V}_{\text{TM}}$ with at most one node selected from each, and then edges to $\mathcal{V}_{\text{Post}}$ with a maximum of 3 nodes. The weight function $\omega : \mathcal{E} \rightarrow [0, 1]^k$ assigns only normalized parameters to edges, where k denotes the total number of continuous parameters across connected nodes. For an edge $e = (u, v) \in \mathcal{E}$, $\omega(e) = [\hat{\theta}_u; \hat{\theta}_v]$ with $\hat{\theta}_u = (\theta_u - \theta_u^{\min}) / (\theta_u^{\max} - \theta_u^{\min})$ representing the normalized parameter of node u , and $\hat{\theta}_v$ defined analogously for node v . This structure captures algorithm-specific operations through typed node sets and their normalized parameters via edge weights, while preserving the mandatory sequence of RAW-to-RGB conversion.

Following the graph-based definition of operation chains, we introduce a GNN f_{GNN} to extract structural features from $\mathcal{Q}$. That aligns the feature of Q_j with the trace $\mathbf{F}_j$ in a shared space. Specifically, the f_{GNN} processes node attributes and edge weights through message-passing layers GraphConv [65], where nodes in $\mathcal{V}$ are initialized with one-hot encodings reflecting their operation-algorithm identities and edges utilize normalized parameters $\omega(e) \in [0, 1]^k$ as weights to modulate information propagation. After propagation steps, a graph-level feature $\mathbf{Q}_j$ is obtained via global pooling, denoted as $\mathbf{Q}_j = f_{\text{GNN}}(Q_j) \in \mathbb{R}^d$ with the same dimension as $\mathbf{F}_j$. To enforce alignment between the paired $\{(\mathbf{Q}_j, \mathbf{F}_j)\}_{j \in \mathcal{A}}$ in a $|\mathcal{A}|$ -sample mini-batch, we implement a multimodal contrastive loss $\mathcal{L}_{\text{RAW-Graph}}$ inspired by the symmetric cross entropy loss of CLIP [43]:

$$\mathcal{L}_{\text{cx}}(\mathbf{Q}, \mathbf{F}) = -\frac{1}{|\mathcal{A}|} \sum_{j \in \mathcal{A}} \log \frac{\exp(\mathbf{Q}_j \cdot \mathbf{F}_j / \tau)}{\sum_{l \in \mathcal{A}} \exp(\mathbf{Q}_j \cdot \mathbf{F}_l / \tau)}, \quad (10)$$

where τ is a temperature. Then $\mathcal{L}_{\text{RAW-Graph}}$ is derived as:

$$\mathcal{L}_{\text{RAW-Graph}} = \frac{1}{2} (\mathcal{L}_{\text{cx}}(\mathbf{Q}, \mathbf{F}) + \mathcal{L}_{\text{cx}}(\mathbf{F}, \mathbf{G})) \quad (11)$$

3.4. RAW-Vision Branch

In addition to the topological perspective of the RAW signal, we also supervise the learning of this alignment space from its visual perspective. Notably, traditional pixel-level supervision (*e.g.*, L1/L2 losses) has critical limitations [30, 70]. It focuses solely on pixel-wise error metrics, fails to capture the high-order feature distributions of images, and is

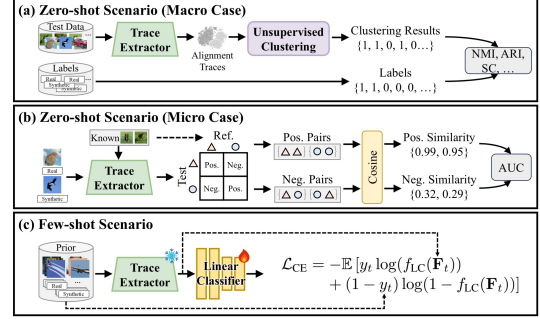


Figure 4. Evaluation process in zero-/few-shot scenarios.

overly sensitive to noise and minor spatial shifts [29, 31]. Moreover, it tends to prioritize local pixel matching over global semantic alignment [32]. To tackle these issues, we propose constraining via feature distribution transfer in the latent space.

Specifically, we employ a pre-trained Variational Autoencoder (VAE) [26, 47] whose encoder $\text{Enc}(\cdot)$ maps inputs to latent Gaussian parameters. For a RAW image $\mathbf{R}_i$ and its RGB counterpart $\mathbf{I}_{i,j}$, the encoder outputs:

$$\text{Enc}(\mathbf{R}_i) = \mathcal{N}(\mu_s, \sigma_s), \quad \text{Enc}(\mathbf{I}_{i,j}) = \mathcal{N}(\mu_t, \sigma_t), \quad (12)$$

where μ_s, σ_s denote the mean and variance of $\mathbf{R}_i$'s latent distribution, and μ_t, σ_t correspond to $\mathbf{I}_{i,j}$'s.

Our goal now is to align RAW distribution $\mathcal{N}(\mu_s, \sigma_s)$ with its RGB counterpart $\mathcal{N}(\mu_t, \sigma_t)$ in latent space, guided by $\mathbf{F}_j$. To enable more effective guidance, we employ cross attention between the $\mathbf{F}_j$ and the distribution parameters. Simple operations like dot product or element-wise addition fail to capture the nuanced contextual dependencies as they treat all feature dimensions uniformly, ignoring the varying relevance of different spectrum signals in $\mathbf{F}_j$ to specific elements of the distribution parameters. Formally, we modulate the mean μ_s through cross-attention conditioning on $\mathbf{F}_j$:

$$\hat{\mu}_s = \text{Attn}(\mu_s, \mathbf{F}_j, \mathbf{F}_j). \quad (13)$$

With the modulated distribution, we employ the KL divergence $\mathcal{D}_{\text{KL}}$ to measure their similarity, defining the constraint loss from the visual perspective as:

$$\mathcal{L}_{\text{RAW-Vision}} = \mathbb{E}_{\mathbf{R}_i, \mathbf{I}_{i,j}} [\mathcal{D}_{\text{KL}}(\mathcal{N}(\hat{\mu}_s, \sigma_s) \parallel \mathcal{N}(\mu_t, \sigma_t))], \quad (14)$$

where the KL divergence is calculated as:

$$\begin{aligned} \mathcal{D}_{\text{KL}}(\mathcal{N}(\hat{\mu}_s, \sigma_s) \parallel \mathcal{N}(\mu_t, \sigma_t)) \\ = \frac{1}{2} \left(\frac{(\hat{\mu}_s - \mu_t)^2}{\sigma_t^2} + \frac{\sigma_s^2}{\sigma_t^2} - \log \frac{\sigma_s^2}{\sigma_t^2} - 1 \right). \end{aligned} \quad (15)$$

To comprehensively align RAW signals with RGB-derived trace features from both topological and visual perspectives, we combine the two branch-specific losses as:

$$\mathcal{L}_{\text{total}} = \lambda_1 \cdot \mathcal{L}_{\text{RAW-Graph}} + \lambda_2 \cdot \mathcal{L}_{\text{RAW-Vision}}. \quad (16)$$

Table 1. Evaluations on micro zero-shot scenario via feature clustering.

Synthesis	R2LCM [57]			ReRAW [8]			ZeroFake [49]			Aero [46]			ZED [16]			FSD [39]			MIB [9]			Ours		
	NMI	ARI	SC	NMI	ARI	SC	NMI	ARI	SC	NMI	ARI	SC	NMI	ARI	SC	NMI	ARI	SC	NMI	ARI	SC	NMI	ARI	SC
GANs																								
Star [14]	.382	.540	.411	.496	.602	.561	.671	.695	.762	.158	.193	.278	.964	.729	.784	.986	.998	.831	.947	.974	.731	.999	.999	.894
CRN [13]	.353	.497	.471	.460	.564	.546	.618	.688	.724	.184	.209	.242	.973	.699	.788	.974	.963	.708	.977	.990	.698	.994	.998	.879
IMLE [34]	.161	.242	.214	.108	.142	.155	.126	.214	.169	.209	.273	.599	.342	.417	.435	.605	.712	.558	.701	.774	.587	.981	.992	.631
SITD [12]	.091	.162	.124	.001	.001	.009	.012	.028	.016	.359	.434	.616	.457	.546	.561	.342	.477	.401	.794	.844	.700	.999	.999	.725
SAN [18]	.126	.229	.168	.017	.022	.030	.073	.135	.099	.017	.023	.127	.163	.254	.213	.354	.413	.428	.415	.481	.478	.994	.998	.876
DMs																								
Glide [40]	.109	.205	.147	.077	.105	.147	.379	.408	.407	.060	.081	.109	.487	.592	.530	.612	.666	.669	.802	.881	.674	.953	.976	.871
Dalle [44]	.145	.247	.193	.106	.140	.188	.588	.637	.548	.395	.487	.440	.594	.625	.639	.651	.694	.667	.841	.908	.684	.994	.998	.897
SD [6]	.057	.112	.077	.034	.042	.066	.446	.482	.522	.347	.430	.399	.326	.415	.396	.314	.401	.448	.499	.573	.520	.904	.945	.815
MJ [3]	.002	.006	.003	.000	.000	.004	.217	.282	.285	.557	.622	.620	.467	.551	.544	.472	.547	.587	.730	.806	.678	.990	.996	.883
FireFly [1]	.039	.082	.054	.021	.028	.040	.202	.276	.250	.032	.044	.069	.667	.652	.724	.653	.689	.668	.714	.806	.629	.833	.882	.817
Avg.	.147	.232	.186	.132	.165	.175	.333	.385	.378	.232	.280	.350	.544	.548	.561	.596	.656	.597	.742	.804	.638	.964	.978	.829

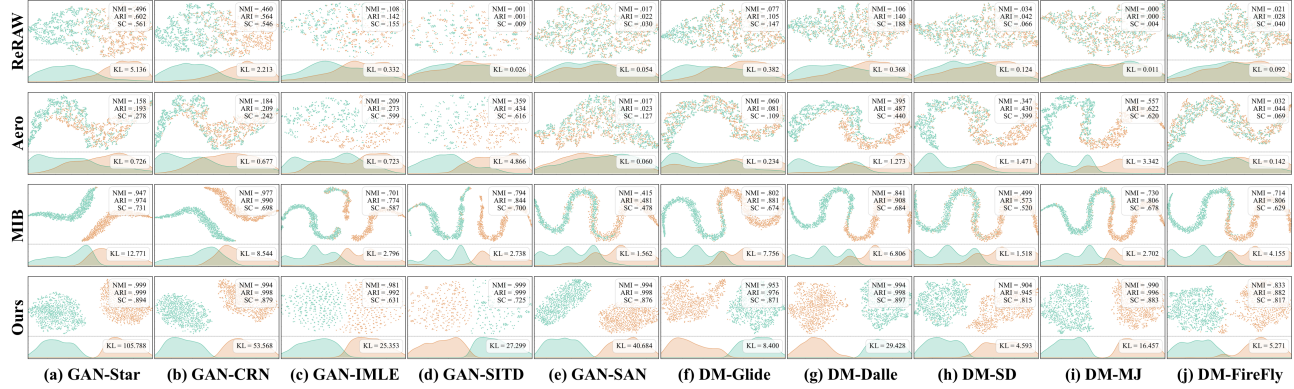


Figure 5. Visualizations of feature clustering and distribution in micro zero-shot scenario.

By jointly minimizing this objective, our model learns to preserve both the structural logic of RAW-to-RGB conversion via $\mathcal{L}_{\text{RAW-Graph}}$ and the statistical characteristics of visual features via $\mathcal{L}_{\text{RAW-Vision}}$ in the shared embedding space.

4. Experiments

4.1. Settings

Scenarios. We evaluate zero-shot and few-shot scenarios as shown in Fig. 4. The former distinguishes synthetic from real images without GenAI priors, using feature clustering or similarity evaluation. The latter investigates whether partial priors enable cross-domain detection, specifically testing on unseen GenAI after training on known ones.

Dataset. During the training phase, the proposed method solely relies on authentic RAW and RGB images from FiveK [10], obviating the need for introducing priors like synthetic images or generation algorithms. FiveK offers 5,000 RAW images captured by Canon and NIKON in DNG format, which can be easily rendered into RGB format using tools like PyRAW and DCRAW. For zero-shot evaluation, we incorporate GANs [12–14, 18, 34] and DMs [40, 44, 47] from widely-used benchmarks [1, 3, 55]; whereas, in few-shot evaluation, we adhere to existing works [41, 52, 55, 68],

employing solely ProGAN [25] synthesis for fine-tuning and subsequently conducting cross-domain detection on the remaining unknown syntheses.

Competitors. For the zero-shot scenario, we select open-source competitors (ZeroFake [49], Aero [46], ZED [16], FSD [39], and MIB [9]) and RAW-reconstruction baselines (R2LCM [57] and ReRAW [8]) for comparison. For the few-shot scenario, we include state-of-the-art competitors: CNNSpot [55], UniDet [41], NPR [52], and Effort [68].

Metrics. For the macro zero-shot scenario, we use NMI, Adjusted Rand Index (ARI), and Silhouette Coefficient (SC) [5]; while for the micro zero-shot scenario, we adopt cosine similarity between positive/negative pairs and AUC [4]. In the few-shot scenario, we involve the AP following [55]. Detailed implementations are provided in Appx. C.

4.2. Zero-shot Scenario: Feature Clustering

Table 1 presents the NMI, ARI, and SC results of different competitors for clustering GAN- and DM-generated images. First, R2LCM and ReRAW, which are not designed for the synthetic detection, show some effectiveness in GAN-Star clustering (NMI: 0.382 and 0.496) but fail for other generations (*e.g.*, NMI drops to 0.057 and 0.034 for DM-SD). Methods tailored for the synthetic detection achieve better

Table 2. Evaluations on macro zero-shot scenario via feature similarity.

Synthesis	R2LCM [57]			ReRAW [8]			ZeroFake [49]			Aero [46]			ZED [16]			FSD [39]			MIB [9]			Ours		
	Neg.	Pos.	AUC	Neg.	Pos.	AUC	Neg.	Pos.	AUC	Neg.	Pos.	AUC	Neg.	Pos.	AUC	Neg.	Pos.	AUC	Neg.	Pos.	AUC	Neg.	Pos.	AUC
GANs																								
Star [14]	.673	.778	.714	.823	.929	.764	.652	.747	.724	.899	.905	.607	.839	.964	.677	.443	.561	.736	.850	.947	.788	.667	.869	.978
CRN [13]	.676	.779	.711	.827	.930	.904	.849	.952	.642	.882	.903	.587	.641	.729	.740	.439	.563	.696	.851	.954	.759	.732	.862	.927
IMLE [34]	.680	.703	.527	.831	.857	.538	.658	.742	.578	.885	.895	.844	.639	.760	.701	.539	.661	.884	.850	.949	.879	.090	.672	.989
SITD [12]	.716	.741	.539	.869	.889	.556	.647	.735	.636	.873	.891	.852	.839	.958	.901	.842	.961	.682	.849	.955	.850	.046	.712	.997
SAN [18]	.715	.730	.519	.861	.880	.545	.825	.946	.599	.879	.898	.567	.635	.758	.754	.438	.562	.723	.850	.951	.749	.644	.813	.917
DMs																								
Glide [40]	.689	.711	.528	.837	.859	.580	.843	.954	.614	.894	.895	.545	.738	.852	.683	.539	.652	.643	.852	.948	.703	.223	.587	.872
Dalle [44]	.697	.717	.527	.843	.862	.568	.656	.750	.639	.888	.894	.719	.637	.762	.658	.644	.756	.716	.850	.948	.704	.119	.579	.928
SD [6]	.676	.702	.532	.826	.856	.558	.865	.943	.591	.890	.892	.579	.642	.760	.702	.579	.620	.566	.882	.918	.582	.249	.600	.848
MJ [3]	.668	.688	.524	.813	.832	.544	.744	.852	.744	.893	.895	.576	.736	.860	.688	.639	.716	.614	.859	.946	.729	.275	.646	.932
FireFly [1]	.674	.697	.533	.825	.850	.564	.747	.853	.637	.893	.894	.537	.482	.522	.564	.880	.923	.573	.881	.917	.593	.209	.558	.859
Avg.	.686	.725	.565	.836	.874	.612	.749	.847	.640	.888	.896	.641	.683	.793	.707	.598	.698	.683	.857	.943	.734	.325	.690	.925

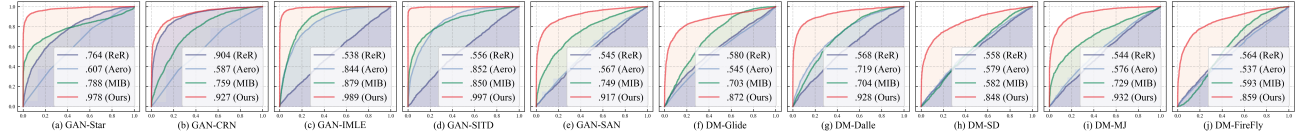


Figure 6. Visualizations of ROC curves in macro zero-shot scenario.

results. For instance, ZED and FSD attain NMI values of 0.964 and 0.986 respectively on GAN-Star. Nevertheless, due to the lack of explicit consideration of the differences between synthetic and real images in the physical space, they struggle to achieve good discriminability across more types of GANs and DMs, with an average NMI of only 0.544 and 0.596. In contrast, by characterizing the RAW-RGB joint alignment space, our method can better explore the intrinsic differences between synthetic and real images. As a result, the extracted alignment trace exhibits stronger generalization, achieving an average NMI of 0.964, corresponding to a 22.2% improvement over the second-ranked method. Additionally, Fig. 5 visualizes the feature spaces using T-SNE for scatter plots and distribution maps. Obviously, our method can achieve more discriminative feature clustering, *e.g.*, it gets a KL value of 29.4 in DM-Dalle, far exceeding the competitors' values of 6.8, 1.2, and 0.3.

4.3. Zero-shot Scenario: Feature Similarity

From a micro perspective, Table 2 evaluates the ability of different forensic features to distinguish whether two images belong to the same “Synthetic” or “Real” category. Ideally, the feature similarity of positive pairs should be much higher than that of negative pairs, and this statistical characteristic is reflected in a higher AUC value, where an AUC of 0.5 indicates random guessing. Obviously, the features extracted by many competitors fail to achieve ideal discriminability. For instance, when targeting DM-generated images, the similarity differences between positive and negative pairs of ReRAW and Aero are both within 10%, resulting in most of their final AUC values being below 0.60. In contrast, our method achieves a similarity difference of over 34.9% across

Table 3. Detection results on ForenSynths [55] (AP criterion).

Forensics	Generative Adversarial				Perceptual		Low-level		Facial	Avg.
	Style	Big	Cycle	Star	Gau	CRN	IMLE	SITD	SAN	Deepfake
CNNSpot [55]	.996	.845	.935	.982	.895	.982	.984	.972	.705	.890
CR [11]	.986	.881	.957	.939	.990	.993	.998	.799	.657	.716
DIRE [59]	.985	.942	.917	.919	.960	.933	.891	.835	.696	.780
HiFi [20]	.964	.921	.910	.958	.955	.826	.898	.902	.649	.738
UniDet [41]	.972	.996	.995	.996	.999	.967	.990	.613	.790	.825
NPR [52]	.998	.945	.998	.999	.999	.926	.974	.910	.765	.902
Effort [68]	.950	.996	.998	.999	.996	.989	.989	.925	.815	.876
Ours	.999	.998	.999	.999	.999	.997	.999	.984	.957	.940

all DM-generated images, and reaches an average AUC of 0.925 that far exceeds the second-best's 0.734. Fig. 6 also visualizes the ROC curves and AUC values, which clearly demonstrate the significant advantage of our method.

4.4. Few-shot Scenario: Synthetic Detection

Although this work focuses on zero-shot detection, the RAW-RGB joint feature space, constructed based on the inherent properties of physical signals, provides a solid forensic foundation. When partial GenAI prior data is available, this well-initialized feature space can be fine-tuned to yield far stronger generalization in discriminative capabilities compared to suboptimal starting points. Table 3 presents the detection performance on ten cross-domain GenAIs in the ForenSynths [55], with only ProGAN used as prior data for training/fine-tuning. As observed, most methods can generalize to detect GenAIs of the same type, *e.g.*, achieving near 0.999 AP on StyleGAN, StarGAN, and GauGAN. But their performance generally degrades when dealing with GenAIs of greater divergence, *e.g.*, CNNSpot and Effort only reach 0.705 and 0.815 AP respectively when detecting SAN. In contrast, our method not only generalizes well from ProGAN to similar generative adversarial types but also adapts

Table 4. Impact of RGB-Vision branch variants.

#	LVM	Freeze	Attn.	Evaluation		
				Cluster	Similarity	Detection
#1	ConvNeXt	✗	✗	.863	.810	.882
#2	EffNet	✗	✗	.854	.783	.862
#3	EffNet	✓	SEA	.891	.833	.912
#4	SegFormer	✗	✗	.792	.814	.875
#5	ViT	✗	✗	.867	.830	.909
#6	ViT	✓	✗	.908	.882	.963
#7	ViT	✓	Cross	.929	.892	.969
#8	ViT	✓	SEA	.964	.925	.987

effectively to perceptual, low-level, and facial ones. Ultimately, we achieve an average AP of 0.987, outperforming the second-ranked method by 3.4%.

4.5. Ablation Studies

Tables 4–6 validate the gains of the three proposed branches, where each table’s last row shows the optimal variant. We adopt the aforementioned cluster, similarity, and detection evaluations with NMI, AUC, and AP metrics, respectively.

RGB-Vision Branch Table 4 presents different pre-trained LVM backbones and attention strategies used to extract alignment traces from RGB signals. While our method exhibits generality and versatility, it achieves optimal performance on the ViT backbone compared to ConvNeXt [37], EfficientNet [53], and SegFormer [66]. However, allowing all parameters to be trainable compromises the prior knowledge learned from large-scale data, *e.g.*, #5 shows a 4.1% drop in NMI and 5.2% drop in AUC compared to #6 regarding the learning of general forensic features [41]. Finally, the comparison between #7 (standard cross-attention [54]) and #8 (our proposed SEA) demonstrates that the latter better captures camera internal parameters potentially retained in RGB signals from the ISP process, yielding 3.5% higher NMI and 3.3% higher AUC.

RAW-Graph Branch Table 5 demonstrates the impact of different RGB signals and topology graphs derived from RAW signals. First, #1–#3 clearly illustrate the importance of the number of pipelines P_j : only when they include both ISP and post-processing can they adequately cover diverse RAW-to-RGB transformations, effectively enhancing the generalization of subsequent alignment traces. Second, using a simple undirected topology graph (#4) or failing to assign weights to edge nodes (#5) leads to inaccurate graph construction, causing confusion between different trace types and hindering subsequent optimization. Finally, comparing different loss functions in #6–#8 shows that InfoNCE [43] outperforms Triplet [7] or Circle [51] in optimizing cross-modal comparisons between RGB and RAW representations, achieving the best performance across all three evaluations.

Table 5. Impact of RAW-Graph branch variants.

#	P_j	Represent	Loss	Evaluation		
				Cluster	Similarity	Detection
#1	ISP	Text	InfoNCE	.836	.790	.843
#2	Post	Text	InfoNCE	.827	.816	.858
#3	All	Text	InfoNCE	.868	.847	.885
#4	All	Topol. U	InfoNCE	.909	.882	.917
#5	All	Topol. D	InfoNCE	.939	.910	.958
#6	All	Topol. D+W	Triplet	.892	.853	.924
#7	All	Topol. D+W	Circle	.951	.894	.965
#8	All	Topol. D+W	InfoNCE	.964	.925	.987

Table 6. Impact of RAW-Vision branch variants.

#	Encoder	Loss	Evaluation			
			Recon.	Cluster	Similarity	Detection
#1	✗	✗	✗	.828	.817	.896
#2	UNet	L2	.721	.876	.853	.908
#3	VAE	L2	.786	.912	.873	.915
#4	UNet	KL	.754	.905	.884	.912
#5	VAE	KL	.866	.938	.917	.975
#6	VAE	KL+Attn.	.898	.964	.925	.987

RAW-Vision Branch Table 6 illustrates the impact of RAW signals on modeling alignment traces from a visual perspective. Here, we additionally introduce the SSIM [58] metric to evaluate reconstructive performance, where higher values indicate smaller reconstruction errors. #1 shows that removing this branch significantly limits the ability to fully capture the representational diversity of RAW signals, leading to substantial performance degradation. This is because topology graphs are abstract, higher-level concepts that can map to multiple instantiated RGB signals; thus, dual constraints from a visual perspective are necessary. #2–#5 compare different autoencoder paradigms, including architectures and training approaches focused on either pixel- or latent-level. Notably, VAE+KL (#5) proves more effective by using a partial differential training paradigm with KL divergence imposing constraints on feature distributions. It improves the SSIM by 14.5% compared to the UNet+L2 (#2), which in turn drives significant gains across clustering, similarity, and detection metrics. Finally, #6 demonstrates that incorporating an attention strategy further enhances the interaction between alignment traces and RAW visual features, resulting in moderate performance improvements.

5. Conclusions

This work reconsiders the distinction between synthetic and real images as rooted in the physical world, and reveals that reconstructed RAW signals of digital images serve as a powerful discriminative tool. We thereby propose characterizing a RAW–RGB joint alignment space optimized through three branches: RGB vision, RAW graph, and RAW vision. Experiments verify that it achieves promising performance in zero-shot settings and can be efficiently applied to cross-domain detection with partial GenAI priors.

Acknowledgment

This work was supported by the National Natural Science Foundation of China under U25A20429, 62332018, and W2511073; by Macau Science and Technology Development Fund under 001/2024/SKL, 0119/2024/RIB2, and 0110/2025/RIB2; by Research Committee at University of Macau under MYRG-CRG2025-00031-FST and MYRG-GRG2025-00086-FST; by the Guangdong Basic and Applied Basic Research Foundation under Grant 2024A1515012536; also by TeleAI under QNZS-20251008.

References

- [1] Adobe firefly. <https://firefly.adobe.com/>. 6, 7
- [2] Libraw library. <https://www.libraw.org>. 3, 1
- [3] Midjourney: Expanding the imaginative powers. <https://midjourney.com/>. 6, 7
- [4] Area under the roc curve. https://scikit-learn.org/stable/modules/generated/sklearn.metrics.roc_auc_score.html. 6
- [5] Clustering performance evaluation. <https://scikit-learn.org/stable/modules/clustering.html>. 6
- [6] Stability.ai: Stable diffusion public release. <https://stability.ai/blog/stable-diffusion-public-release>. 6, 7
- [7] V. Balntas, E. Riba, D. Ponsa, and K. Mikolajczyk. Learning local feature descriptors with triplets and shallow convolutional neural networks. In *Proc. British Mach. Vis. Conf.*, pages 1–11, 2016. 8
- [8] R. Berdan, B. Besbinar, C. Reinders, J. Otsuka, and D. Iso. Reraw: Rgb-to-raw image reconstruction via stratified sampling for efficient object detection on the edge. In *Proc. Comput. Vis. Pattern Recogn.*, pages 11833–11843, 2025. 2, 3, 6, 7
- [9] J. Brokman, A. Giloni, O. Hofman, R. Vainshtein, H. Kojima, and G. Gilboa. Manifold induced biases for zero-shot and few-shot detection of generated images. In *Proc. Int. Conf. Learn. Represent.*, pages 1–26, 2025. 2, 6, 7
- [10] V. Bychkovsky, S. Paris, E. Chan, and F. Durand. Learning photographic global tonal adjustment with a database of input/output image pairs. In *Proc. Comput. Vis. Pattern Recogn.*, pages 97–104, 2011. 6
- [11] K. Chandrasegaran, N. T. Tran, A. Binder, and N. M. Cheung. Discovering transferable forensic features for cnn-generated images detection. In *Proc. Eur. Conf. Comput. Vis.*, pages 671–689, 2022. 2, 7
- [12] C. Chen, Q. Chen, J. Xu, and V. Koltun. Learning to see in the dark. In *Proc. Comput. Vis. Pattern Recogn.*, pages 3291–3300, 2018. 6, 7
- [13] Q. Chen and V. Koltun. Photographic image synthesis with cascaded refinement networks. In *Proc. Int. Conf. Comput. Vis.*, pages 1511–1520, 2017. 6, 7
- [14] Y. Choi, M. Choi, M. Kim, J. Ha, S. Kim, and J. Choo. Stargan: Unified generative adversarial networks for multi-domain image-to-image translation. In *Proc. Comput. Vis. Pattern Recogn.*, pages 8789–8797, 2018. 6, 7
- [15] H. Ci, Z. Peng, P. Yang, Y. Xuan, and M. Shou. Diffseg30k: A multi-turn diffusion editing benchmark for localized aige detection. *arXiv preprint arXiv:2511.19111*, 2025. 2
- [16] D. Cozzolino, G. Poggi, M. Nießner, and L. Verdoliva. Zero-shot detection of ai-generated images. In *Proc. Eur. Conf. Comput. Vis.*, pages 54–72, 2024. 2, 6, 7
- [17] W. Cui, D. Wang, and N. Han. Survey on fake information generation, dissemination and detection. *Chin. J. Electron.*, 33(3):573–583, 2024. 1
- [18] T. Dai, J. Cai, Y. Zhang, S. Xia, and L. Zhang. Second-order attention network for single image super-resolution. In *Proc. Comput. Vis. Pattern Recogn.*, pages 11065–11074, 2019. 6, 7
- [19] I. Goodfellow, J. Pouget-Abadie, M. Mirza, B. Xu, D. Warde-Farley, S. Ozair, A. Courville, and Y. Bengio. Generative adversarial nets. In *Proc. Neural Info. Process. Syst.*, pages 2672–2680, 2014. 1
- [20] X. Guo, X. Liu, Z. Ren, S. Grosz, I. Masi, and X. Liu. Hierarchical fine-grained image forgery detection and localization. In *Proc. Comput. Vis. Pattern Recogn.*, pages 3155–3165, 2023. 2, 7
- [21] Z. He, P. Chen, and T. Ho. Rigid: A training-free and model-agnostic framework for robust ai-generated image detection. *arXiv preprint arXiv:2405.20112*, 2024. 2
- [22] J. Ho, A. Jain, and P. Abbeel. Denoising diffusion probabilistic models. In *Adv. in Neur. Info. Proc. Syst.*, pages 6840–6851, 2020. 1
- [23] L. Ilya and H. Frank. Decoupled weight decay regularization. In *Proc. Int. Conf. Learn. Represent.*, 2019. 2
- [24] N. Kanopoulos, N. Vasanthavada, and R. Baker. Design of an image edge detection filter using the sobel operator. *IEEE J. Solid-State Circuits.*, 23(2):358–367, 1988. 3
- [25] T. Karras, T. Aila, S. Laine, and J. Lehtinen. Progressive growing of gans for improved quality, stability, and variation. In *Proc. Int. Conf. Learn. Represent.*, 2018. 6
- [26] D. Kingma and M. Welling. Auto-encoding variational bayes. In *Proc. Int. Conf. Learn. Representat.*, pages 1–14, 2013. 2, 5
- [27] C. Kong, A. Luo, P. Bao, Y. Yu, H. Li, Z. Zheng, S. Wang, and A. Kot. Moe-ffd: Mixture of experts for generalized and parameter-efficient face forgery detection. *IEEE Trans. Dependable Secure Comput.*, pages 1–15, 2025. 1
- [28] C. Kong, A. Luo, S. Wang, H. Li, A. Rocha, and A. Kot. Pixel-inconsistency modeling for image manipulation localization. *IEEE Trans. Pattern Anal. and Mach. Intell.*, 47(6):4455–4472, 2025. 1
- [29] F. Li, K. Li, J. Tian, and J. Zhou. Regroup median loss for combating label noise. In *Proc. AAAI Conf. Arti. Intell.*, pages 13474–13482, 2024. 5
- [30] F. Li, K. Li, H. Wu, J. Tian, and J. Zhou. DAT: Improving adversarial robustness via generative amplitude mix-up in frequency domain. In *Proc. Adv. Neural Inf. Process. Syst.*, pages 127099–127128, 2024. 5
- [31] F. Li, K. Li, Q. Wang, B. Han, J. Tian, and J. Zhou. RML++: regroup median loss for combating label noise. *Int. J. Comput. Vis.*, 133(9):6400–6421, 2025. 5

- [32] F. Li, K. Li, H. Wu, J. Tian, and J. Zhou. Toward robust learning via core feature-aware adversarial training. *IEEE Trans. Inf. Forensics Secur.*, 20:6236–6251, 2025. 5
- [33] F. Li, K. Li, Q. Wang, B. Han, and J. Zhou. AEGIS: Adversarial target-guided retention-data-free robust concept erasure from diffusion models. In *Proc. Int. Conf. Learn. Represent.*, 2026. 1
- [34] K. Li, T. Zhang, and J. Malik. Diverse image synthesis from semantic layouts via conditional imle. In *Proc. Int. Conf. Comput. Vis.*, pages 4220–4229, 2019. 6, 7
- [35] K. Li, Q. Wang, Y. Wang, F. Li, J. Liu, B. Han, and J. Zhou. LLM unlearning with LLM beliefs. In *Proc. Int. Conf. Learn. Represent.*, 2026. 1
- [36] Z. Li, M. Lu, X. Zhang, X. Feng, M. Asif, and Z. Ma. Efficient visual computing with camera raw snapshots. *IEEE Trans. Pattern Anal. and Mach. Intell.*, 46(7):4684–4701, 2024. 2
- [37] Z. Liu, H. Mao, C. Wu, C. Feichtenhofer, T. Darrell, and S. Xie. A convnet for the 2020s. In *Proc. Comput. Vis. Pattern Recogn.*, pages 11976–11986, 2022. 8
- [38] Y. Luo, J. Du, K. Yan, and S. Ding. Lare²: Latent reconstruction error based method for diffusion-generated image detection. In *Proc. Comput. Vis. Pattern Recogn.*, pages 17006–17015, 2024. 1
- [39] T. Nguyen, A. Azizpour, and M. Stamm. Forensic self-descriptions are all you need for zero-shot detection, open-set source attribution, and clustering of ai-generated images. In *Proc. Comput. Vis. Pattern Recogn.*, pages 3040–3050, 2025. 2, 6, 7
- [40] A. Nichol, P. Dhariwal, A. Ramesh, P. Shyam, P. Mishkin, B. McGrew, I. Sutskever, and M. Chen. Glide: Towards photorealistic image generation and editing with text-guided diffusion models. In *Proc. Int. Conf. Mach. Learn.*, pages 16784–16804, 2022. 6, 7
- [41] U. Ojha, Y. Li, and Y. Lee. Towards universal fake image detectors that generalize across generative models. In *Proc. Comput. Vis. Pattern Recogn.*, pages 24480–24489, 2023. 1, 2, 6, 7, 8
- [42] M. Oquab, T. Darcet, T. Moutakanni, H. Vo, M. Szafraniec, V. Khalidov, P. Fernandez, D. Haziza, F. Massa, A. El-Nouby, et al. Dinov2: Learning robust visual features without supervision. *arXiv preprint arXiv:2304.07193*, 2023. 2
- [43] A. Radford, J. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, K. Gretchen, and S. Ilya. Learning transferable visual models from natural language supervision. In *Proc. Int. Conf. Mach. Learn.*, pages 8748–8763, 2021. 5, 8, 2
- [44] A. Ramesh, P. Dhariwal, A. Nichol, C. Chu, and M. Chen. Hierarchical text-conditional image generation with clip latents. *preprint arXiv:2204.06125*, 2022. 6, 7
- [45] C. Reinders, R. Berdan, B. Besbinar, J. Otsuka, and D. Iso. Raw-diffusion: Rgb-guided diffusion models for high-fidelity raw image generation. In *Proc. IEEE Winter Conf. Appl. Comput. Vis.*, pages 8431–8443, 2025. 2
- [46] J. Ricker, D. Lukovnikov, and A. Fischer. Aeroblade: Training-free detection of latent diffusion images using autoencoder reconstruction error. In *Proc. Comput. Vis. Pattern Recogn.*, pages 9130–9140, 2024. 6, 7
- [47] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, and B. Ommer. High-resolution image synthesis with latent diffusion models. In *CVPR*, pages 10684–10695, 2022. 5, 6
- [48] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, and B. Ommer. Stable diffusion. <https://github.com/CompVis/stable-diffusion>, 2022. 2
- [49] Z. Sha, Y. Tan, M. Li, M. Backes, and Y. Zhang. Zerofake: Zero-shot detection of fake images generated and edited by text-to-image generation models. In *Proc. ACM Conf. Comput. Commun. Secur.*, pages 4852–4866, 2024. 2, 6, 7
- [50] C. Shannon. A mathematical theory of communication. *Bell Syst. Tech. J.*, 27(3):379–423, 1948. 4
- [51] Y. Sun, C. Cheng, Y. Zhang, C. Zhang, L. Zheng, Z. Wang, and Y. Wei. Circle loss: A unified perspective of pair similarity optimization. In *Proc. Comput. Vis. Pattern Recogn.*, pages 6398–6407, 2020. 8
- [52] C. Tan, Y. Zhao, S. Wei, G. Gu, P. Liu, and Y. Wei. Rethinking the up-sampling operations in cnn-based generative network for generalizable deepfake detection. In *Proc. Comput. Vis. Pattern Recogn.*, pages 28130–28139, 2024. 1, 2, 6, 7
- [53] M. Tan and Q. Le. Efficientnet: Rethinking model scaling for convolutional neural networks. In *Proc. Int. Conf. Mach. Learn.*, pages 6105–6114, 2019. 8
- [54] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. Gomez, L. Kaiser, and I. Polosukhin. Attention is all you need. In *Proc. Adv. Neural Inf. Process. Syst.*, pages 1–11, 2017. 4, 8
- [55] S. Wang, O. Wang, R. Zhang, A. Owens, and A. A. Efros. Cnn-generated images are surprisingly easy to spot... for now. In *Proc. Comput. Vis. Pattern Recogn.*, pages 8695–8704, 2020. 1, 2, 6, 7
- [56] X. Wang, K. Yu, S. Wu, J. Gu, Y. Liu, C. Dong, Y. Qiao, and C. Loy. Esrgan: Enhanced super-resolution generative adversarial networks. In *Proc. Eur. Conf. Comput. Vis. Workshop*, pages 1–16, 2018. 2
- [57] Y. Wang, Y. Yu, W. Yang, L. Guo, L. Chau, A. Kot, and B. Wen. Beyond learned metadata-based raw image reconstruction. *Int. J. Comput. Vis.*, 132(12):5514–5533, 2024. 6, 7
- [58] Z. Wang, A. Bovik, H. Sheikh, and E. Simoncelli. Image quality assessment: from error visibility to structural similarity. *IEEE Trans. Image Process.*, 13(4):600–612, 2004. 8
- [59] Z. Wang, J. Bao, W. Zhou, W. Wang, H. Hu, H. Chen, and H. Li. Dire for diffusion-generated image detection. In *Proc. Int. Conf. Comput. Vis.*, pages 22445–22455, 2023. 2, 7
- [60] K. Wong, J. Zhou, K. Li, Y. Si, X. Wu, and J. Zhou. FontGuard: A robust font watermarking approach leveraging deep font knowledge. *IEEE Trans. Multimedia*, 27:7876–7890, 2025. 1
- [61] K. Wong, J. Zhou, H. Wu, Y. Si, and J. Zhou. Adcd-net: Robust document image forgery localization via adaptive dct feature and hierarchical content disentanglement. In *Proc. Int. Conf. Comput. Vis.*, pages 19280–19289, 2025. 1
- [62] H. Wu, J. Zhou, X. Zhang, J. Tian, and W. Sun. Robust camera model identification over online social network shared images via multi-scenario learning. *IEEE Trans. Inf. Forensics Secur.*, 19(1):148–162, 2024. 4

- [63] H. Wu, Y. Chen, J. Zhou, and Y. Li. Rethinking image forgery detection via soft contrastive learning and unsupervised clustering. *IEEE Trans. Dependable Secure Comput.*, 22(6):6296–6308, 2025. [1](#)
- [64] H. Wu, J. Zhou, and S. Zhang. Generalizable synthetic image detection via language-guided contrastive learning. *IEEE Trans. on Artificial Intelligence*, pages 1–11, 2025. [1](#), [2](#)
- [65] Z. Wu, S. Pan, F. Chen, G. Long, C. Zhang, and P. Yu. A comprehensive survey on graph neural networks. *IEEE Trans. Neural Netw. Learn. Syst.*, 32(1):4–24, 2020. [4](#), [5](#)
- [66] E. Xie, W. Wang, Z. Yu, A. Anandkumar, J. Alvarez, and P. Luo. Segformer: Simple and efficient design for semantic segmentation with transformers. In *Proc. Neural Info. Process. Syst.*, pages 12077–12090, 2021. [8](#)
- [67] Y. Xing, Z. Qian, and Q. Chen. Invertible image signal processing. In *Proc. Comput. Vis. Pattern Recogn.*, pages 6287–6296, 2021. [2](#)
- [68] Z. Yan, J. Wang, P. Jin, K. Zhang, C. Liu, S. Chen, T. Yao, S. Ding, B. Wu, and L. Yuan. Orthogonal subspace decomposition for generalizable ai-generated image detection. In *Proc. Int. Conf. Mach. Learn.*, pages 1–21, 2025. [1](#), [2](#), [6](#), [7](#)
- [69] Y. Yu, W. Yang, Y. Tan, and A. Kot. Towards robust rain removal against adversarial attacks: A comprehensive benchmark analysis and beyond. In *Proc. Comput. Vis. Pattern Recogn.*, pages 6013–6022, 2022. [3](#)
- [70] Y. Yu, Y. Wang, W. Yang, L. Guo, S. Lu, L. Duan, Y. Tan, and A. Kot. Robust and transferable backdoor attacks against deep image compression with selective frequency prior. *IEEE Trans. Pattern Anal. Mach. Intell.*, 47(3):1674–1693, 2025. [5](#)
- [71] B. Zheng, N. Ma, S. Tong, and S. Xie. Diffusion transformers with representation autoencoders. *arXiv preprint arXiv:2510.11690*, 2025. [2](#)
- [72] C. Zheng, A. Shrivastava, and A. Owens. Exif as language: Learning cross-modal associations between images and camera metadata. In *Proc. Comput. Vis. Pattern Recogn.*, pages 6945–6956, 2023. [4](#)